QUASI-COMPACTNESS DESCENDS ALONG A UNIVERSALLY OPEN SURJECTIVE MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a descent argument on points and open covers; the local quasi-compactness of algebraic spaces is the only topological input, and the rest is the quasi-compactness criterion.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Quasi-compactness of a morphism of algebraic spaces tested after base change from quasi-compact spaces. Lemma lemma-characterize-representable-quasi-compact, used at proof lines 1360-1361 and 1372-1373. Is the criterion the proof verifies.
- Points of an algebraic space have fundamental systems of quasi-compact open neighbourhoods. Properties of Spaces, Lemma spaces-properties-lemma-space-locally-quasi-compact, used at proof lines 1362-1367. Allows a quasi-compact open W surjecting onto Z to be found inside Y' x_Y Z.
- A surjective image of a quasi-compact space is quasi-compact. Lemma lemma-surjection-from-quasi-compact, used at proof lines 1370-1373. Transfers quasi-compactness from W x_Y X to Z x_Y X.
- Openness and surjectivity are stable under base change. used at proof lines 1361-1362. Makes Y' x_Y Z -> Z open and surjective.

Source reference. Original lemma 040W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5271–5271. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6086–6086. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $g : Y' \to Y$ be a universally open and surjective morphism of algebraic spaces such that the base change $f' : X' \to Y'$ is quasi-compact. Then f is quasi-compact.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $Z \to Y$ be a morphism of algebraic spaces with Z quasi-compact. As g is universally open and surjective, we see that $Y' \times_Y Z \to Z$ is open and surjective. As every point of $|Y' \times_Y Z|$ has a fundamental system of quasi-compact open neighbourhoods (see Properties of Spaces, Lemma 04NN) we can find a quasi-compact open $W \subset |Y' \times_Y Z|$ which surjects onto Z. Denote $f'' : W \times_Y X \to W$ the base change of f' by $W \to Y'$. By assumption $W \times_Y X$ is quasi-compact. As $W \to Z$ is surjective we see that $W \times_Y X \to Z \times_Y X$ is surjective. Hence $Z \times_Y X$ is quasi-compact by Lemma 040W. Thus f is quasi-compact. $\square$